

Sudipta Das

AI/ML Engineer | Python | PyTorch | LLM Applications | Computer Vision

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Summary

AI/ML Engineer with hands-on experience building Python applications and developing machine learning and deep learning models. Project experience covers LLM applications, customer churn prediction, and multimodal computer vision, with practical skills in data preprocessing, model evaluation, and API integration. Interested in applying AI to agriculture and intelligent decision-support systems that address real-world needs.

Technical Skills

Programming: Python, SQL

Machine Learning: scikit-learn, XGBoost, Random Forest, SVM, classification, regression, model evaluation

Deep Learning: PyTorch, CNN, transfer learning, model fine-tuning

NLP & LLM: Gemini API, Sentence Transformers, text embeddings, semantic search, FAISS

Computer Vision: Image preprocessing, CLIP, multimodal learning, image-text retrieval

Data: NumPy, Pandas, Matplotlib, Seaborn, feature engineering, data preprocessing

Development: REST API integration, Streamlit, Requests, GitHub, ML inference pipelines

Projects

CineMaya AI | AI-Powered Movie Recommendation System

- Built an AI-powered recommendation application with Streamlit, Gemini API, Sentence Transformers and FAISS, combining LLM query enhancement with embedding-based semantic search.
- Integrated TMDb metadata via REST APIs with rating-based ranking, caching, and API failure fallback.

GitHub: github.com/diptadaswork21-code/CineMaya-AI

Customer Churn Prediction AI Platform | 2026

- Developed an end-to-end machine learning pipeline using XGBoost and scikit-learn, covering data preprocessing, feature engineering, model training and evaluation.
- Built a Streamlit interface for model inference, customer churn prediction and data visualization.

GitHub: github.com/diptadaswork21-code/customer-churn-ai

Current Project

AI Research Paper Assistant | RAG-Based LLM Application (Ongoing)

- Developing a Retrieval-Augmented Generation (RAG) assistant in Python for research paper analysis, with document processing, text chunking, text embeddings, and semantic search.
- Building FastAPI REST APIs for document ingestion and context-aware question answering as part of NLP and AI application development.
- Exploring Sentence Transformers, FAISS, and vector database options alongside LLM APIs for summarization and knowledge extraction using Large Language Models (LLMs).

Experience

Brain Station 23 | Industrial Attachment in Machine Learning | 2025 (2 weeks)

- Worked on Python-based data preprocessing, ML model development and evaluation; gained practical exposure to machine learning pipelines and ML deployment concepts.

Research Experience

Multimodal AI System for Misinformation Detection Using CLIP | 2025-2026

- Developed a computer vision and multimodal AI system for image-text misinformation detection using CLIP embeddings, multimodal preprocessing, model fine-tuning and evaluation.
- Tested domain adaptation and cross-domain generalization; documented methods and findings.

Education

Chittagong University of Engineering and Technology (CUET)

M.Sc. (Engg.), Electronics and Telecommunication Engineering | 2026-Present

B.Sc. (Engg.), Electronics and Telecommunication Engineering | 2022-2026 | CGPA: 3.49/4.00